Huanzhi Pu

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Seeking: Summer 2027 research internship in GPU architecture, GPU/ML compilers, HW/SW co-design, or performance analysis

Education

- **Georgia Institute of Technology** | Ph.D. in Computer Science — Expected May 2030
  M.S. in Computer Science; Concentration: Computing Systems; GPA: 3.92/4.0 — May 2026
  B.S. in Computer Science; Concentrations: Theory, Systems & Architecture; GPA: 3.93/4.0; Dean's List — May 2024

Research Experience

- **Georgia Tech HPArch Lab** — Atlanta, GA
  Graduate Research Assistant / Undergraduate Researcher | Advisor: Prof. Hyesoon Kim — Aug 2022 – Present
  - **CUDA on Non-NVIDIA Platforms:** Extended CuPBoP, an open-source toolchain that compiles CUDA programs for CPU and RISC-V GPU targets, adding support for cooperative groups, warp-level primitives, and atomics.
  - **CuLifter: Lifting GPU Binaries to Typed IR:** Led development of a binary translator that lifts NVIDIA GPU binaries to typed LLVM IR; built a multi-stage pipeline that reconstructs SIMT control flow, restores register types, and normalizes architecture-specific instruction patterns; accepted to MICRO 2026.
  - **VortexSynth: Profile-Guided Benchmark Reconstruction:** Built an LLM agent that migrates robotics GPU workloads to Vortex without source access, constructing a robotics kernel benchmark suite for GPU architecture research.
- **Zhang Research Group, Cornell University** — Remote
  Research Assistant | Advisor: Prof. Zhiru Zhang — May 2023 – May 2025
  - **Sparse Kernels:** Implemented sparse formats and CUDA SpMV kernels; benchmarked them on SuiteSparse.

Publications

- Jisheng Zhao*, **Huanzhi Pu***, Shinnung Jeong, Chihyo Ahn, Hyesoon Kim. "CuLifter: Lifting GPU Binaries to Typed IR." *MICRO 2026*. *Co-first authors.
- Shinnung Jeong, Chihyo Ahn, **Huanzhi Pu**, Jisheng Zhao, Hyesoon Kim, Blaise Tine. "Inside VOLT: Designing an Open-Source GPU Compiler." *CC 2026* (tool paper).
- Yuxiao Jia, Euijun Chung, **Huanzhi Pu**, Ben Feinberg, Hyesoon Kim. "TensorDynamic: Bridging Application- and Instruction-Level Fault Injection for DNN Tensor Core Execution." *ISPASS 2026*.
- Chihyo Ahn, Shinnung Jeong, Liam Paul Cooper, Ruobing Han, **Huanzhi Pu**, Nicholas Parnenzini, Jisheng Zhao, Blaise Tine, Hyesoon Kim. "FastTrackGPU: Static-Analysis–Guided Analytical Modeling for Softcore GPUs." *IEEE Computer Architecture Letters (CAL)*, vol. 25, no. 1, pp. 162–165, 2026.
- **Huanzhi Pu**, Rishabh Ravi, Shinnung Jeong, Udit Subramanya, Euijun Chung, Jisheng Zhao, Chihyo Ahn, Hyesoon Kim. "Hardware vs. Software Implementation of Warp-Level Features in Vortex RISC-V GPU." *DATE 2025 Workshop (OSSMPIC)*.
- Jie Liu, **Huanzhi Pu**, Zhiru Zhang. "AsyncSparse: Accelerating Sparse Matrix-Matrix Multiplication on Asynchronous GPU Architectures." *arXiv preprint*, 2026.
- **Huanzhi Pu***, Chulhyung Park*, Chihyo Ahn, Ramyad Hadidi, Hyesoon Kim. "VortexSynth: Profile-guided benchmark reconstruction for GPU architecture research." *arXiv preprint*, 2026. *Co-first authors.

Teaching

- **Georgia Institute of Technology** | Teaching Assistant, CS 8803: GPU Software and Hardware — 2023 – 2026 (5 semesters)
  - Maintained a SASS→LLVM lifter for a thread-divergence course project; held office hours; advised on CUDA optimization.

Technical Skills

Languages: C/C++, CUDA, Python, PTX/SASS

Compilers & Tools: LLVM, GPU binary analysis, PyTorch, cuSPARSE, Linux, Git, Docker